Project Title

Develop a computer network analysis application using a binary tree to calculate its diameter and maximum width**.**

1. Problem Statement

To develop a Computer Network Analysis Application using a Binary Tree in C++ to calculate the diameter and maximum width of a network represented as a binary tree. The diameter represents the longest path between any two nodes, and the maximum width represents the maximum number of nodes present at any level.

2. Objectives

1. To understand the concept of a Binary Tree.
2. To represent a computer network using a binary tree.
3. To calculate the diameter of the binary tree.
4. To calculate the maximum width of the binary tree.
5. To display the analysis results.
6. To understand the practical application of binary trees.

3. Introduction

A Binary Tree is a non-linear data structure in which each node can have a maximum of two children: a left child and a right child.

In this project, a binary tree represents a computer network structure. Each node represents a network device, and the edges represent connections between devices.

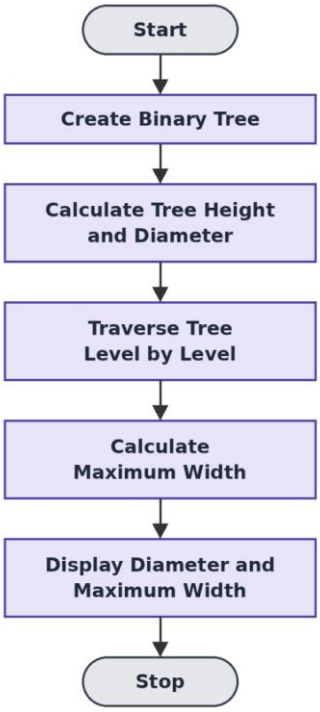
The diameter of a binary tree is the number of edges in the longest path between any two nodes. The maximum width is the largest number of nodes present at any level of the tree.

This project uses C++ to calculate the diameter and maximum width of the binary tree and display the results.

4. Algorithm

1. Start the program.
2. Create a binary tree using nodes.
3. Assign data to each node and connect the left and right children.
4. Calculate the height of the left and right subtrees.
5. Calculate the diameter by finding the longest path between any two nodes.
6. Traverse the tree level by level using a queue.
7. Calculate the number of nodes at each level.
8. Find and display the maximum width.
9. Display the diameter and maximum width of the network.
10. Stop the program.

5. Flowchart



6. Result

The Computer Network Analysis Application was implemented successfully using a Binary Tree in C++. The program calculates the diameter as 4 edges and the maximum width as 3 nodes for the sample tree.

7. Conclusion

This project helped us understand the implementation of a Binary Tree in C++. It demonstrates how tree traversal and height calculation can be used to determine the diameter and maximum width of a tree. The project provides a basic understanding of representing network structures using tree data structures.